

Part X: Rheumatology and Immunology

Chapter 10A - Rheumatology Overview: Joint Pain, Inflammation, Autoantibodies, Vasculitis, and Immunosuppression

Chapter purpose and clinical frame

This chapter is a diagnostic scaffold for later rheumatology chapters. Rheumatology is often introduced as a collection of named autoimmune diseases, but the safer clinical approach is to begin with patterns: articular versus periarticular pain, inflammatory versus mechanical pain, acute versus chronic tempo, monoarticular versus oligoarticular versus polyarticular distribution, peripheral versus axial disease, and localized versus systemic involvement. A patient does not present with "rheumatoid arthritis" or "vasculitis"; the patient presents with a hot knee, morning stiffness, hand swelling, rash, kidney injury, hemoptysis, neuropathy, visual symptoms, or unexplained inflammatory markers. The task is to transform those clues into a problem representation that determines what must be ruled out immediately and what can be evaluated systematically.

The highest-yield rule is that rheumatology is pattern recognition plus danger recognition. Pattern recognition prevents wasteful testing. Danger recognition prevents catastrophe. Broad serology without a compatible phenotype produces false-positive ANA, RF, and ANCA results that distract from the real diagnosis. Conversely, a patient with hematuria, pulmonary hemorrhage, vision loss, mononeuritis multiplex, or digital ischemia may need treatment before every confirmatory test has returned. For graph-RAG extraction, the repeated association should be: joint pattern + inflammatory features + extra-articular organ clues + targeted testing + treatment risk.

Rheumatologic red flags: hot swollen joint; fever with effusion; new visual symptoms, jaw claudication, or scalp tenderness; pulmonary hemorrhage with renal injury; rapidly progressive glomerulonephritis; CNS disease; mononeuritis multiplex; digital ischemia; threatened airway; severe cytopenias; severe proximal weakness with dysphagia or respiratory symptoms; and immunosuppressed fever.

Approach to joint pain

The first question is whether the pain is truly articular. True arthritis implies pathology inside the joint space: synovitis, effusion, cartilage damage, crystal deposition, infection, or hemarthrosis. It tends to produce swelling, warmth, pain with both active and passive movement, and loss of range of motion in multiple planes. Periarticular disorders, including bursitis, tendinopathy, enthesitis, tenosynovitis, rotator cuff disease, plantar fasciitis, and trochanteric pain syndrome, often hurt with selected active movements but have relatively preserved passive motion and lack a true effusion. Bone pain, fracture, osteonecrosis, malignancy, and neuropathic pain can all be described by patients as "joint pain," but they do not behave like synovitis.

The second question is how many joints are involved. Acute monoarthritis is treated as septic arthritis until proven otherwise when an effusion and inflammatory signs are present. Acute oligoarthritis suggests infection, crystal disease, reactive arthritis, early spondyloarthritis, Lyme disease in endemic areas, inflammatory bowel disease-associated arthritis, or gonococcal disease. Chronic symmetric polyarthritis points toward rheumatoid arthritis, systemic lupus erythematosus, viral arthritis, polyarticular osteoarthritis, endocrine/metabolic disease, or chronic inflammatory arthritis. Axial pain with inflammatory features suggests axial spondyloarthritis. The pattern is often more informative than any single antibody.

Table: Initial joint pain questions

Question	What to look for	Interpretation	Common trap
Articular or periarticular?	Effusion, warmth, passive ROM loss, pain in multiple planes	True arthritis suggests septic arthritis, crystal disease, RA, SLE, OA flare, hemarthrosis. Periarticular pain suggests tendon, bursa, enthesitis, ligament, or nerve source.	Calling tendon or bursa pain "arthritis" and ordering broad serology.
How many joints?	One; 2-4; five or more; axial involvement	Monoarthritis: septic, crystal, trauma. Oligoarthritis: reactive arthritis, spondyloarthritis, Lyme, gonococcal disease. Polyarthritis: RA, SLE, viral, OA, endocrine.	Ignoring a single hot joint because the patient also has chronic OA.
How fast did it start?	Hours-days, weeks-months, years	Hours-days suggests infection, crystal, hemarthrosis, trauma. Weeks suggests inflammatory arthritis or viral disease. Years suggests OA, chronic RA, spondyloarthritis, or mechanical disease.	Assuming chronic disease excludes superimposed infection.
Inflammatory or mechanical?	Morning stiffness, rest pain, night pain, swelling, activity response	Inflammatory pain improves with gentle activity and has prolonged stiffness. Mechanical pain worsens with use and improves with rest.	Older adults can develop inflammatory disease; young adults can have mechanical injury.

Question	What to look for	Interpretation	Common trap
Systemic disease?	Fever, weight loss, rash, ulcers, Raynaud, sicca, nephritis, neuropathy, eye disease, bowel disease	Systemic clues convert joint pain into connective tissue disease, vasculitis, infection, malignancy, endocrine disease, or drug toxicity.	Treating pain alone while missing kidney, lung, eye, nerve, or vascular disease.

Monoarticular joint pain

Acutely painful monoarthritis has a narrow emergency differential: septic arthritis, crystal arthritis, hemarthrosis, fracture/internal derangement, and less commonly Lyme arthritis, gonococcal arthritis, reactive arthritis, or the first presentation of systemic inflammatory disease. The diagnostic reflex should be arthrocentesis when an effusion is present and infection is plausible. Serum uric acid, radiographs, ESR, CRP, and peripheral WBC count do not rule out septic arthritis. Crystals do not rule out infection because crystal disease and infection can coexist in the same joint. Antibiotics should generally be given after blood cultures and synovial fluid are obtained, unless the patient is unstable.

The distribution of monoarthritis matters. A first metatarsophalangeal attack with abrupt severe pain favors gout, but a hot knee, wrist, shoulder, ankle, elbow, or hip demands broader evaluation. Hip and shoulder septic arthritis can be subtle because deep joints may not show obvious swelling. Prosthetic joints are high-risk because biofilm infection may present indolently, inflammatory markers may be misleading, and source control often requires procedural management.

Polyarticular joint pain

Polyarticular pain should be mapped, not merely counted. Symmetric small-joint synovitis of the MCP, PIP, wrists, and MTPs suggests rheumatoid arthritis. DIP-predominant disease suggests osteoarthritis or psoriatic arthritis. Migratory arthritis suggests rheumatic fever, disseminated gonococcal infection, Lyme disease, viral infection, endocarditis, or SLE. Additive arthritis suggests RA or another chronic inflammatory arthritis. Axial pain, enthesitis, dactylitis, uveitis, psoriasis, and inflammatory bowel disease point toward spondyloarthritis. Rash, photosensitivity, oral ulcers, pleurisy, nephritis, cytopenias, and low complement point toward SLE or immune-complex disease.

Polyarthritis is not automatically autoimmune. Viral arthritis can mimic RA, especially parvovirus B19, hepatitis B, hepatitis C, HIV, chikungunya, and other travel-related alphaviruses. Endocarditis can cause arthralgia, immune-complex glomerulonephritis, low complement, and positive rheumatoid factor. Hemochromatosis, thyroid disease, hyperparathyroidism, sarcoidosis, malignancy, and medication-associated syndromes can also cause joint symptoms. Immune checkpoint inhibitors can cause inflammatory arthritis; aromatase inhibitors cause arthralgia; retinoids can produce musculoskeletal pain; statins can produce myopathy that may be mistaken for diffuse rheumatic pain.

Table: Pattern recognition in joint pain

Pattern	High-yield differential	Clues that move the diagnosis
Acute monoarthritis	Septic arthritis, gout, CPPD, hemarthrosis, fracture/internal derangement	Fever, bacteremia risk, prosthetic joint, immunosuppression, crystals, trauma, anticoagulation.
Acute oligoarthritis	Reactive arthritis, gonococcal arthritis, Lyme arthritis, viral arthritis, spondyloarthritis, crystal disease	Urethritis/cervicitis, diarrhea, tick exposure, migratory tenosynovitis, uveitis, enthesitis.
Chronic symmetric small-joint polyarthritis	RA, SLE, viral arthritis, psoriatic arthritis, polyarticular OA	MCP/PIP/wrist swelling, anti-CCP, RF, erosions, rash, cytopenias, sicca, psoriasis.
Axial inflammatory pain	Axial spondyloarthritis, psoriatic arthritis, IBD-associated arthritis	Back stiffness improves with exercise, alternating buttock pain, uveitis, HLA-B27 context, sacroiliitis.
Migratory arthritis	Rheumatic fever, disseminated gonococcal infection, Lyme, endocarditis, SLE	Sequential joints, fever, murmur, rash, tenosynovitis, exposure history.
DIP-predominant hand pain	OA, psoriatic arthritis, gout	Heberden nodes, nail pitting, dactylitis, erosive OA, tophi.

Inflammatory versus noninflammatory arthritis

Inflammatory arthritis reflects immune activation, infection, or crystal-driven innate immune inflammation inside the joint. The classic history is morning stiffness lasting more than 30-60 minutes, stiffness after inactivity, pain at rest or at night, warmth, swelling, fatigue, and improvement with gentle movement. Noninflammatory arthritis is usually mechanical, degenerative, traumatic, or structural. It worsens with use, improves with rest, has brief morning stiffness, and shows bony enlargement or crepitus more than soft synovial swelling. The distinction is not perfect: severe OA can flare with effusion, crystal disease can occur in damaged OA joints, and chronic inflammatory arthritis can create secondary mechanical pain after joint destruction.

Physical examination is the best discriminator. Synovitis feels soft, boggy, and sometimes warm; bony hypertrophy feels hard; tendon sheath inflammation tracks along a tendon; enthesitis localizes to insertion points; bursitis produces focal extra-articular swelling. In the hands, RA commonly involves MCP and PIP joints and usually spares DIP joints, while OA commonly involves DIP, PIP, and first CMC joints. Inflammatory back pain often begins before age 40, improves with exercise, does not improve with rest, and can awaken the patient in the second half of the night.

Table: Inflammatory versus mechanical arthritis

Feature	Inflammatory arthritis	Noninflammatory/mechanical arthritis
Morning stiffness	Often prolonged, especially more than 30-60 minutes; stiffness after rest is prominent.	Usually brief, often less than 30 minutes; loosens quickly.
Pain behavior	May hurt at rest or overnight; improves with gentle activity.	Worse with use, load, repetitive motion; improves with rest.
Exam	Warmth, effusion, soft boggy synovial swelling, limited passive ROM.	Bony enlargement, crepitus, malalignment, focal tendon/bursa tenderness.
Systemic features	Fatigue, fever, weight loss, rash, eye disease, mucosal ulcers, serositis can occur.	Usually absent unless another disease process is present.
Inflammatory markers	Can be elevated, but normal values do not exclude disease.	Usually normal unless injury, OA flare, infection, crystal disease, or comorbidity.
Synovial fluid	Inflammatory WBC count; often neutrophil-predominant in acute disease.	Noninflammatory WBC count; viscous fluid; crystals absent unless comorbid crystal disease.
Examples	RA, SLE arthritis, psoriatic arthritis, reactive arthritis, gout, CPPD, septic arthritis.	OA, tendinopathy, bursitis, hypermobility, trauma, avascular necrosis.

Acute arthritis emergencies

Septic arthritis

Septic arthritis is the must-not-miss diagnosis in acute inflammatory monoarthritis. Risk factors include bacteremia, older age, diabetes, chronic kidney disease, immunosuppression, skin infection, injection drug use, recent joint surgery, prosthetic joint, rheumatoid arthritis, and preexisting joint damage. *Staphylococcus aureus* is the commonest cause in many adult settings; *Neisseria gonorrhoeae* should be considered in sexually active patients with migratory polyarthralgia, tenosynovitis, dermatitis, or purulent monoarthritis. Gram-negative organisms are more likely with urinary or gastrointestinal sources, older age, immunocompromise, or health-care exposure.

The workup of a hot swollen joint is not complete without synovial fluid if fluid can be safely obtained. Send cell count with differential, Gram stain, aerobic and anaerobic culture, and crystal analysis. Blood cultures are important because bacteremia may identify the organism even when synovial Gram stain is negative. Imaging can locate deep effusions or adjacent osteomyelitis, but imaging does not replace aspiration. Crystals, high serum urate, or prior gout history must not prematurely close the case.

Septic arthritis management is urgent source control plus antibiotics. Aspirate or surgically drain the joint, start empiric antibiotics after cultures when possible, tailor therapy to culture results, and involve orthopedics for hip, shoulder, prosthetic joint, failed aspiration, loculated infection, or rapid deterioration. Delay can cause irreversible cartilage damage, osteomyelitis, sepsis, and disability.

Crystal arthritis

Crystal arthritis can look as inflammatory as infection. Gout results from monosodium urate crystals and classically causes abrupt severe pain, swelling, erythema, and disability. CPPD disease results from calcium pyrophosphate crystals and often affects older adults, knees, wrists, shoulders, ankles, and MCP joints; it may be triggered by illness, surgery, trauma, or metabolic disorders such as hemochromatosis, hyperparathyroidism, hypomagnesemia, or hypothyroidism. Definitive diagnosis is crystal identification under polarized microscopy, but the presence of crystals should not stop culture if infection is plausible.

Acute therapy for crystal arthritis depends on kidney function, infection concern, anticoagulation, diabetes, and comorbidities. NSAIDs, colchicine, and glucocorticoids are common options, but intra-articular steroid should be avoided until infection is reasonably excluded. Serum urate can be normal during an acute gout flare, so a normal value does not rule out gout. Urate-lowering therapy decisions belong to long-term management, not the initial emergency distinction between septic and nonseptic arthritis.

Inflammatory flare with systemic danger

Known autoimmune disease does not make every flare benign. A patient with SLE and joint pain may also have nephritis, CNS lupus, pericarditis, myocarditis, pneumonitis, pulmonary embolism from antiphospholipid syndrome, or infection from immunosuppression. A patient with RA may have septic arthritis in a damaged joint, interstitial lung disease, pericarditis, vasculitis, or drug toxicity. When the presentation includes fever, hypoxia, hemoptysis, renal injury, neurologic deficits, severe cytopenias, visual symptoms, or digital ischemia, the task is not only to treat pain but to identify organ-threatening disease.

Table: Acute arthritis and systemic emergencies

Emergency pattern	Immediate concern	Initial actions	Pearl
Hot swollen single joint	Septic arthritis until proven otherwise	Arthrocentesis; blood cultures; synovial cell count, Gram stain, culture, crystals; empiric antibiotics after cultures if stable.	Crystals do not exclude infection.
Acute arthritis plus purpura, dermatitis, tenosynovitis, or sexual exposure	Disseminated gonococcal infection	NAAT testing at exposed sites; blood/synovial cultures; ceftriaxone-based therapy per local guidance.	Migratory symptoms can precede purulent arthritis.
Acute arthritis plus anticoagulation or trauma	Hemarthrosis, fracture, internal derangement	Assess stability; imaging; aspirate if tense/painful or infection uncertain.	Bloody fluid can be traumatic tap or true hemarthrosis.
Arthritis plus hemoptysis and AKI/hematuria	Pulmonary-renal syndrome	Urinalysis, creatinine, CBC, chest imaging, ANCA, anti-GBM, complements; urgent rheum/neph/pulm.	Do not wait for every serology when life-threatening hemorrhage is suspected.
Headache plus jaw claudication or vision symptoms in older adult	Giant cell arteritis with threatened vision	High-dose glucocorticoids immediately; ESR/CRP; temporal artery imaging/biopsy; ophthalmology/rheum.	Treat before biopsy if vision is at risk.
Digital ischemia, livedo, ulcers, mononeuritis multiplex	Vasculitis, emboli, antiphospholipid syndrome, cryoglobulinemia	Vascular exam, pulses, labs, urine, vascular imaging, antiphospholipid/cryoglobulin testing when appropriate.	Think vessel size and organ threat.

Morning stiffness, systemic symptoms, and extra-articular clues

Morning stiffness is a time-based proxy for synovial inflammation, but it must be interpreted with the whole picture. Prolonged stiffness after sleep or inactivity supports inflammatory arthritis, whereas brief stiffness after starting movement is common in OA. Fibromyalgia can produce subjective stiffness without objective synovitis, and severe OA may produce longer discomfort. The diagnostic question is therefore not only "How long are you stiff?" but also "Are the joints swollen? Are they warm? Does gentle movement help? Are there systemic or extra-articular clues?"

Systemic symptoms raise the stakes. Fever can signal infection, Still disease, vasculitis, malignancy, drug reaction, or uncontrolled connective tissue disease. Weight loss and night sweats can occur in vasculitis and autoimmune disease but should also trigger evaluation for infection and malignancy. Fatigue is common and nonspecific; it becomes diagnostically useful when paired with objective inflammation, cytopenias, rash, nephritis, serositis, or elevated inflammatory markers.

Table: Extra-articular clues

Clue	Associated diagnoses	How it changes evaluation
Photosensitive rash, oral/nasal ulcers, alopecia	SLE, dermatomyositis, drug reactions	ANA-first approach only if phenotype fits; CBC, urinalysis, complements, anti-dsDNA if SLE likely.
Raynaud phenomenon	Systemic sclerosis, mixed connective tissue disease, SLE, myositis overlap, primary Raynaud	Ask about digital ulcers, puffy fingers, dysphagia, dyspnea; consider ANA/ENA guided by exam.
Sicca symptoms	Sjogren syndrome, RA-associated sicca, medications, hepatitis C, sarcoidosis	Dryness alone is nonspecific; look for parotid enlargement, dental caries, neuropathy, renal tubular acidosis.
Uveitis or painful red eye	Spondyloarthritis, Behcet, sarcoidosis, vasculitis, infection	Urgent ophthalmology if vision/pain/photophobia; ask about back pain, ulcers, IBD, psoriasis.
Psoriasis, nail pitting, dactylitis	Psoriatic arthritis	DIP disease and enthesitis help separate from RA and OA.
Diarrhea, urethritis, cervicitis	Reactive arthritis, IBD-associated arthritis, gonococcal infection	Obtain exposure history; test for STIs when appropriate; examine entheses and eyes.
Hematuria/proteinuria	Lupus nephritis, ANCA vasculitis, IgA vasculitis, anti-GBM, infection-related GN	Urinalysis is a rheumatology vital sign; nephritic sediment can be organ-threatening.
Hemoptysis/dyspnea	Pulmonary hemorrhage, ILD, PE, serositis, infection, drug toxicity	Chest imaging, oxygenation, CBC, renal tests, urinalysis; urgent escalation when pulmonary-renal pattern exists.
Neuropathy, foot drop, wrist drop	Vasculitis, diabetes, amyloid, sarcoid, compression	Mononeuritis multiplex is a vasculitis red flag.
Jaw claudication, scalp tenderness, vision loss	Giant cell arteritis	Start steroids urgently if suspected; do not wait for confirmatory testing if vision threatened.

Laboratory interpretation: use tests to answer a clinical question

Rheumatology labs are often misused as broad screens. The correct use is Bayesian: test when the clinical syndrome gives the result a meaningful pretest probability. ANA positivity can occur in healthy people and in many non-lupus conditions; RF can be positive in chronic infections, older age, lung disease, and other autoimmune diseases; ANCA can be positive in inflammatory bowel disease, infection, drug exposure, and nonvasculitic autoimmune disease. Conversely, negative tests do not always exclude disease when the phenotype is compelling. The lab must be interpreted with organ involvement, exam findings, urinalysis, imaging, and disease tempo.

ESR and CRP

ESR and CRP are inflammation markers, not disease labels. CRP rises and falls more quickly and reflects hepatic acute-phase response, often through IL-6 signaling. ESR is influenced by fibrinogen and immunoglobulins but also by age, sex, anemia, pregnancy, kidney disease, and paraproteins; it can be high without active rheumatic inflammation. Both can be normal in some active diseases, including limited SLE manifestations, some myositis presentations, and early or localized disease. Discordance can be useful: very high ESR with modest CRP may occur in SLE or paraproteinemia, whereas bacterial infection and crystal arthritis often raise CRP substantially.

Table: ESR/CRP interpretation

Pattern	Interpretation	Examples and cautions
High ESR plus high CRP	Active inflammation likely, but not specific.	Infection, crystal arthritis, PMR/GCA, RA flare, vasculitis, malignancy.
High ESR with lower CRP	Immunoglobulin/fibrinogen effects or selected autoimmune patterns.	SLE, anemia, renal disease, pregnancy, monoclonal gammopathy; still evaluate clinically.
High CRP with mild ESR	Rapid acute-phase inflammation or early process.	Bacterial infection, crystal flare, early inflammatory flare.
Normal ESR/CRP	Does not exclude all rheumatic disease.	OA, fibromyalgia, early inflammatory arthritis, some SLE, limited vasculitis, treated disease.
Very high values	Think beyond routine arthritis.	Sepsis, endocarditis, GCA/PMR, vasculitis, malignancy, severe crystal disease.

Autoantibody interpretation

Autoantibodies are pattern amplifiers. A positive ANA in a patient with photosensitive rash, inflammatory arthritis, oral ulcers, cytopenias, nephritic urine, and low complement is meaningful. The same ANA in isolated mechanical knee pain is usually noise. Anti-CCP in symmetric small-joint synovitis strongly supports RA and predicts erosive disease risk. RF supports RA in the right phenotype but is less specific. Anti-dsDNA and anti-Sm are SLE-associated; anti-dsDNA often tracks lupus nephritis activity in context, while anti-Sm is specific but less sensitive. Antiphospholipid antibodies matter when there is thrombosis, pregnancy morbidity, livedo, thrombocytopenia, or lupus context and require persistence over time for classification.

ANCA testing should be ordered when there is a vasculitis phenotype: pulmonary-renal syndrome, rapidly progressive glomerulonephritis, chronic destructive sinusitis with lung lesions, mononeuritis multiplex, palpable purpura with systemic disease, scleritis, or unexplained multisystem inflammation. PR3-ANCA is classically associated with granulomatosis with polyangiitis, while MPO-ANCA is associated with microscopic polyangiitis and eosinophilic granulomatosis with polyangiitis, but overlap and false positives occur. Tissue diagnosis is often needed when treatment would involve high-risk immunosuppression.

Table: Autoantibody interpretation

Test	Best clinical use	Important interpretation points
ANA	Entry test when SLE, Sjogren, systemic sclerosis, myositis overlap, or mixed connective tissue disease is plausible.	Sensitive for SLE but nonspecific. Low-titer positives are common. Do not order for isolated mechanical pain or fatigue without objective clues.
RF	Supports RA when chronic symmetric inflammatory small-joint arthritis is present.	Less specific than anti-CCP. Can be positive in hepatitis C, endocarditis, other autoimmune disease, lung disease, and older adults.
Anti-CCP	RA diagnosis/prognosis in compatible inflammatory arthritis.	More specific for RA than RF. Associated with erosive disease risk. Can precede clinical RA.
Anti-dsDNA	SLE support and nephritis/activity monitoring in context.	Specific for SLE; titers may correlate with nephritis activity in many patients but not all. Interpret with urine, complements, and symptoms.
Anti-Sm	Specific SLE marker.	Low sensitivity; positive result is diagnostically helpful in the right clinical context.

Test	Best clinical use	Important interpretation points
Antiphospholipid antibodies	Thrombosis, recurrent pregnancy morbidity, livedo, thrombocytopenia, lupus context.	Panel includes lupus anticoagulant, anticardiolipin, and anti-beta2 glycoprotein I. Persistent positivity is important; acute illness can cause transient positives.
ANCA with PR3/MPO	Suspected small-vessel vasculitis.	Use when phenotype supports vasculitis. False positives occur. PR3 often GPA; MPO often MPA/EGPA, but clinical context and biopsy matter.
ENA panel	Subclassification after positive ANA and compatible phenotype.	SSA/Ro, SSB/La, RNP, Scl-70, centromere, Jo-1 and related antibodies are useful when the syndrome suggests their disease family.

Complement patterns

Complement levels help identify immune-complex activity. Low C3 and low C4 suggest classical pathway consumption, as in active SLE, cryoglobulinemic vasculitis, infection-related glomerulonephritis, or hypocomplementemic urticarial vasculitis. Low C3 with relatively normal C4 suggests alternative pathway activation, as in some postinfectious or complement-mediated glomerulopathies. Normal complement does not exclude lupus, vasculitis, or inflammatory arthritis, especially if disease is limited or treated. Complement interpretation is most useful when paired with urinalysis, anti-dsDNA, kidney function, rash, neuropathy, pulmonary findings, and infection evaluation.

Table: Complement patterns

Pattern	Mechanistic hint	Common associations
Low C3 plus low C4	Classical pathway immune-complex consumption.	SLE flare/nephritis, cryoglobulinemia, immune-complex vasculitis, infection-related GN, endocarditis-associated immune-complex disease.
Low C3 with normal or less-low C4	Alternative pathway activation or predominant C3 consumption.	Postinfectious GN, C3 glomerulopathy, some infection-related immune-complex states.
Low C4 with normal C3	Classical pathway activation, inherited low C4, or cryoglobulins depending on context.	Cryoglobulinemia, hereditary angioedema evaluation, some lupus contexts.
Normal C3/C4	No major systemic complement consumption at that moment.	Does not rule out RA, spondyloarthritis, limited SLE, ANCA vasculitis, OA, fibromyalgia, crystal disease.

Synovial fluid interpretation

Synovial fluid is a direct sample of the disease compartment. It is particularly important in acute monoarthritis, unexplained effusions, suspected infection, suspected crystal disease, and hemarthrosis. Gross appearance, viscosity, WBC count, differential, Gram stain, culture, and crystal microscopy are interpreted together. The WBC count is helpful but not absolute: septic arthritis and crystal arthritis can both exceed 50,000 cells/mcL, and early or partially treated infection can have lower counts. Immunosuppressed patients may mount less impressive inflammatory responses. Therefore synovial fluid categories should be treated as probability ranges rather than rigid cutoffs.

Table: Synovial fluid categories

Category	Appearance	Approximate WBC pattern	Interpretation
Normal/noninflammatory	Clear, straw-colored, viscous	Usually less than 1,000/mcL	OA, trauma, osteonecrosis, internal derangement. Low count makes infection less likely but does not replace judgment.
Inflammatory	Cloudy, less viscous	Often 2,000-50,000/mcL	RA, SLE, psoriatic arthritis, reactive arthritis, crystal disease. Neutrophils often predominate in acute inflammation.
Highly inflammatory / possible septic	Opaque or purulent	Often greater than 50,000/mcL; sometimes greater than 100,000/mcL	Septic arthritis and crystal disease overlap. Culture and Gram stain are essential. Do not dismiss infection because crystals are found.
Hemorrhagic	Bloody or xanthochromic	Variable, RBC-rich	Trauma, anticoagulation, hemophilia, tumor, neuropathic joint, traumatic tap.
Crystal-positive	Cloudy or inflammatory; crystals on microscopy	Variable; can be very high	Gout: needle-shaped negatively birefringent monosodium urate. CPPD: rhomboid weakly positively birefringent calcium pyrophosphate. Culture if infection plausible.

Practical arthrocentesis algorithm

1. Identify whether an effusion is present. If the joint is deep or uncertain, use ultrasound guidance. 2. Before antibiotics when feasible, obtain synovial fluid for cell count with differential, Gram stain, culture, and crystals; obtain blood cultures if infection is plausible. 3. If purulent fluid, positive Gram stain, bacteremia, systemic toxicity, prosthetic joint, or high clinical concern exists, treat as septic arthritis while awaiting cultures. 4. If crystals are present but infection risk remains, continue culture-based evaluation. 5. If noninflammatory fluid is found, redirect toward OA, trauma, internal derangement, osteonecrosis, or referred pain, but reassess if symptoms worsen.

Never inject glucocorticoid into a joint until infection is reasonably excluded. Intra-articular steroids can temporarily improve pain while worsening an unrecognized infection.

Vasculitis by vessel size

Vasculitis is inflammation of blood vessels, but the clinical expression depends heavily on vessel size and organ distribution. Large-vessel vasculitis causes limb claudication, pulse deficits, bruits, aortic disease, constitutional symptoms, and cranial ischemia in giant cell arteritis. Medium-vessel vasculitis causes organ ischemia, nodules, livedo, ulcers, neuropathy, renal infarcts, hypertension, and aneurysms. Small-vessel vasculitis causes palpable purpura, glomerulonephritis, pulmonary hemorrhage, alveolar infiltrates, sinus/lung disease, mononeuritis multiplex, and capillaritis. The workup should begin with involved organ and vessel-size phenotype rather than a blind antibody panel.

Two conceptual categories are especially useful. ANCA-associated vasculitis is usually pauci-immune small-vessel vasculitis and includes granulomatosis with polyangiitis, microscopic polyangiitis, and eosinophilic granulomatosis with polyangiitis. Immune-complex small-vessel vasculitis is more likely to have low complement and includes IgA vasculitis, cryoglobulinemic vasculitis, lupus vasculitis, hypocomplementemic urticarial vasculitis, and infection-related immune-complex vasculitis. Both categories can threaten kidneys, nerves, lungs, skin, and other organs, but their serologic patterns and associated diseases differ.

Table: Vasculitis by vessel size

Vessel size	Representative diseases	Clinical clues	Testing strategy
Large vessel	Giant cell arteritis, Takayasu arteritis	Limb claudication, pulse deficits, bruits, aortic aneurysm; GCA adds headache, scalp tenderness, jaw claudication, vision loss in older adults.	ESR/CRP, vascular imaging, temporal artery ultrasound/biopsy when GCA suspected; treat vision-threatening GCA immediately.
Medium vessel	Polyarteritis nodosa, Kawasaki disease	Livedo, nodules, ulcers, mononeuritis multiplex, renal infarcts/hypertension, abdominal pain, aneurysms; classic PAN usually spares glomeruli.	Angiography or tissue biopsy; hepatitis B testing when PAN considered; urinalysis helps separate from small-vessel GN.
Small vessel: ANCA-associated	GPA, MPA, EGPA	RPGN, hematuria, pulmonary hemorrhage, sinusitis/otitis, lung nodules, asthma/eosinophilia in EGPA, neuropathy, purpura.	Urinalysis, creatinine, chest imaging, ANCA PR3/MPO, biopsy of kidney/skin/lung/ENT lesion when feasible.
Small vessel: immune complex	IgA vasculitis, cryoglobulinemia, lupus vasculitis, hypocomplementemic urticarial vasculitis, infection-related vasculitis	Palpable purpura, arthralgia, abdominal pain, GN; low complement often in cryoglobulinemia/lupus/infection-related disease.	C3/C4, urinalysis, hepatitis B/C, HIV when appropriate, cryoglobulins, ANA/dsDNA in lupus phenotype, skin/kidney biopsy.
Variable vessel	Behcet disease, relapsing polychondritis, Cogan syndrome	Oral/genital ulcers, uveitis, pathergy, venous/arterial thrombosis; ear/nose/airway cartilage inflammation in relapsing polychondritis.	Clinical diagnosis supported by organ evaluation; exclude infection, IBD, malignancy, and mimics.

Pulmonary-renal syndrome

Pulmonary-renal syndrome is one of the most important rheumatologic emergencies. The pattern is hemoptysis or diffuse alveolar hemorrhage plus nephritic urine sediment, rising creatinine, or rapidly progressive glomerulonephritis. Major causes include ANCA-associated vasculitis, anti-GBM disease, SLE, cryoglobulinemia, and infection-related immune-complex disease. Initial testing includes CBC, creatinine, urinalysis with microscopy, urine protein quantification, chest imaging, ANCA with PR3/MPO, anti-GBM antibodies, ANA, anti-dsDNA, complements, hepatitis/HIV testing when appropriate, and blood cultures if infection is possible. Management often requires urgent multidisciplinary treatment before every test has returned.

Immunosuppressive medication overview

Immunosuppressive therapy is chosen by disease mechanism, target organ, severity, fertility/pregnancy considerations, comorbidities, infection risk, vaccination status, and monitoring burden. The most important division is between symptom-relieving anti-inflammatory drugs, steroid-sparing conventional DMARDs, biologic DMARDs, targeted synthetic DMARDs, and organ-threatening induction agents. Glucocorticoids work quickly but are toxic; disease-modifying drugs are used to control the disease process and minimize steroid exposure.

Before starting significant immunosuppression, clinicians commonly assess tuberculosis risk, hepatitis B and C status, HIV when appropriate, vaccination status, baseline CBC/CMP, pregnancy potential, malignancy history, and disease-specific organ baselines such as urinalysis, chest imaging, pulmonary function testing, ophthalmology exam, or bone density. Medication choice must also anticipate monitoring: methotrexate requires liver and blood count monitoring; hydroxychloroquine requires retinal toxicity surveillance; biologics require infection vigilance; cyclophosphamide requires attention to cytopenias, bladder toxicity, infertility, and malignancy risk.

Table: Immunosuppressive and anti-inflammatory medication overview

Class	Examples	Typical role	Risks and monitoring themes
NSAIDs/colchicine	Naproxen, ibuprofen, indomethacin; colchicine	Symptom control for inflammatory pain, crystal flares, serositis in selected cases.	GI bleeding, kidney injury, BP/heart failure effects; colchicine diarrhea, myopathy, cytopenias, interactions, renal dosing.
Glucocorticoids	Prednisone, methylprednisolone, intra-articular triamcinolone	Rapid control of flares and organ-threatening inflammation; bridge until DMARD works.	Infection, hyperglycemia, hypertension, osteoporosis, avascular necrosis, mood changes, cataracts/glaucoma, adrenal suppression.
Conventional synthetic DMARDs	Methotrexate, hydroxychloroquine, sulfasalazine, leflunomide, azathioprine, mycophenolate	Steroid-sparing control of chronic autoimmune disease.	Drug-specific CBC/CMP monitoring; teratogenicity for methotrexate/leflunomide/mycophenolate; retinal monitoring for hydroxychloroquine.
Biologic DMARDs	TNF inhibitors, abatacept, rituximab, IL-6 inhibitors, IL-17/23 inhibitors, belimumab	Targeted immune pathway control for RA, spondyloarthritis, SLE, vasculitis, psoriasis-related disease.	Serious infection risk; TB/hepatitis screening for selected agents; infusion/injection reactions; disease-specific cautions.
Targeted synthetic DMARDs	JAK inhibitors, apremilast	Oral targeted agents for selected inflammatory arthritides.	Infection, zoster, cytopenias/lipids for JAK inhibitors; thrombosis/MACE warnings in selected populations; GI/weight effects for apremilast.
Cytotoxic/induction agents	Cyclophosphamide, high-dose steroids; rituximab in some induction contexts	Organ- or life-threatening vasculitis, severe systemic disease, lupus nephritis alternatives.	Cytopenias, infertility, malignancy, bladder toxicity for cyclophosphamide; profound infection risk; prophylaxis decisions.
Adjunctive prevention	Vaccines, PJP prophylaxis in selected regimens, calcium/vitamin D/bisphosphonates when indicated	Risk reduction during immunosuppression.	Live vaccine timing, pneumococcal/influenza/zoster strategies, bone protection, latent infection management.

Steroid complications and infection risk

Glucocorticoids are simultaneously rescue drugs and toxicity generators. They rapidly suppress cytokine signaling, vascular inflammation, synovitis, and immune-cell trafficking, which is why they are used for severe flares, vasculitis, GCA, lupus nephritis, pulmonary hemorrhage, and other organ-threatening disease. The same effects impair host defense and wound healing. Infection risk rises with dose, duration, age, diabetes, kidney disease, lung disease, concurrent biologic or cytotoxic therapy, lymphopenia, and prior infection. A patient receiving the equivalent of prednisone 20 mg/day or more for two weeks or longer is generally treated as significantly immunosuppressed for many vaccine and infection-risk decisions.

Steroid toxicity should be anticipated from day one. Short-term effects include insomnia, mood changes, appetite, fluid retention, hypertension, hyperglycemia, and leukocytosis. Intermediate and long-term effects include osteoporosis, fractures, avascular necrosis, cataracts, glaucoma, skin thinning, myopathy, infection, adrenal suppression, Cushingoid habitus, weight gain, peptic ulcer risk when combined with NSAIDs, and cardiovascular/metabolic complications. Steroids can also mask fever and inflammatory signs, making infection harder to recognize.

Table: Steroid toxicity domains

Domain	What to watch	Prevention/mitigation concepts
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Domain	What to watch	Prevention/mitigation concepts
Infection	Pneumonia, zoster, opportunistic infections, reactivation risks, muted fever.	Vaccinate before immunosuppression when possible; screen for TB/hepatitis as indicated; consider PJP prophylaxis in high-risk regimens.
Bone	Osteoporosis, fractures, avascular necrosis.	Use lowest effective dose; calcium/vitamin D when appropriate; weight-bearing activity; DXA/bisphosphonate decisions by risk.
Metabolic/cardiovascular	Hyperglycemia, hypertension, dyslipidemia, weight gain, edema.	Monitor glucose/BP; adjust diabetes therapy; taper when possible.
Neuropsychiatric	Insomnia, mood lability, anxiety, mania/psychosis in susceptible patients.	Morning dosing when feasible; counsel patient; monitor high-dose therapy closely.
Ocular/skin/muscle	Cataracts, glaucoma, skin thinning, bruising, poor wound healing, proximal myopathy.	Ophthalmology in long-term use; fall prevention; minimize chronic dose.
Endocrine	Adrenal suppression after prolonged therapy.	Taper rather than abrupt stop after sustained use; stress-dose considerations in selected patients.

Medication safety patterns

Several safety associations recur in rheumatology. Methotrexate is avoided in pregnancy and significant liver disease and requires folic acid plus CBC/CMP monitoring. Hydroxychloroquine is generally not broadly immunosuppressive but can cause retinal toxicity with long-term exposure, especially at higher cumulative dose or renal impairment. TNF inhibitors can reactivate latent TB and hepatitis B in susceptible patients and can worsen demyelinating disease or heart failure in selected contexts. Rituximab depletes B cells and can impair vaccine responses; hepatitis B reactivation risk must be considered. IL-6 blockade can blunt CRP and fever response, so infection may appear deceptively quiet. JAK inhibitors increase risk of herpes zoster and have important warnings in selected higher-risk patients.

Rheumatologic red flags by organ system

The safest rheumatology framework is to separate painful disease from organ-threatening disease. Joint pain alone may be disabling but is rarely immediately life-threatening; kidney, lung, eye, brain, nerve, blood vessel, or severe hematologic involvement can be. The exam and initial labs should therefore include a search for silent organs: urinalysis for blood/protein/casts, creatinine, CBC for cytopenias, oxygen saturation, pulmonary symptoms, neurologic deficits, skin ischemia/purpura, and ocular symptoms. In systemic rheumatic disease, urinalysis is often as important as the joint exam.

Table: Rheumatologic red flags

Red flag	Why it matters	Initial diagnostic response
Hematuria, proteinuria, RBC casts, rising creatinine	Possible lupus nephritis, ANCA vasculitis, anti-GBM, IgA vasculitis, cryoglobulinemia.	Urgent urine microscopy, urine protein, creatinine, complements, anti-dsDNA/ANCA/anti-GBM as phenotype dictates; nephrology.
Hemoptysis, new hypoxia, diffuse infiltrates	Diffuse alveolar hemorrhage, ILD flare, infection, PE.	CXR/CT, CBC trend, renal tests, urinalysis, ANCA/anti-GBM/SLE labs; urgent pulmonary/rheum.
Vision loss, diplopia, jaw claudication, scalp tenderness	GCA or ocular inflammatory disease may cause irreversible vision loss.	Immediate steroids if GCA suspected; ESR/CRP; ophthalmology/rheumatology; temporal artery evaluation.
Mononeuritis multiplex, wrist/foot drop	Vasculitic nerve ischemia can become permanent.	Neurologic exam, EMG/NCS when stable, vasculitis labs, biopsy target search.
Digital ischemia, ulcers, livedo with pain	Vasculitis, emboli, antiphospholipid syndrome, severe Raynaud/systemic sclerosis.	Vascular exam, pulses, antiphospholipid testing when appropriate, imaging, urgent vascular/rheum if threatened digit.
Severe cytopenias	SLE, macrophage activation syndrome, drug toxicity, marrow disease, infection.	CBC smear, hemolysis labs, ferritin if hyperinflammatory syndrome, medication review, infection evaluation.
Fever on biologics/steroids	Serious infection may be muted or atypical.	Do not assume flare; cultures/imaging as indicated; hold immunosuppression decisions case-by-case.
Severe proximal weakness, dysphagia, respiratory symptoms	Inflammatory myopathy, myasthenic mimic, drug myopathy, rhabdomyolysis.	CK/aldolase, medication review, pulmonary mechanics if respiratory concern, myositis workup.

Diagnostic algorithms for common entry points

Algorithm A: hot swollen joint

Confirm true joint involvement: effusion, warmth, limited passive range of motion. Ask whether the patient is unstable, bacteremic, immunosuppressed, has a prosthetic joint, has overlying cellulitis, or has deep joint involvement. Aspirate if feasible and send cell count/differential, Gram stain, culture, and crystals. Obtain blood cultures if febrile or infection risk is meaningful. If septic arthritis is plausible, start empiric antibiotics after cultures and arrange drainage/source control. If crystals are present and cultures are pending, treat pain while continuing infection evaluation if clinical concern remains. If fluid is noninflammatory, redirect to OA flare, trauma, internal derangement, osteonecrosis, or referred pain.

Algorithm B: chronic small-joint polyarthritis

Document objective synovitis in MCP, PIP, wrist, or MTP joints and count involved joints. Characterize symmetry, duration, morning stiffness, nodules, erosive features, functional limitation, and systemic symptoms. Order RF and anti-CCP when RA phenotype is plausible; consider ESR/CRP and baseline CBC/CMP. Screen for mimics when the presentation is atypical: viral hepatitis, parvovirus, HIV, thyroid disease, hemochromatosis, SLE, psoriatic arthritis, and endocarditis if fever, murmur, or immune-complex signs exist. Treat early inflammatory arthritis as time-sensitive because persistent synovitis drives erosions and disability.

Algorithm C: positive ANA

Ask why the ANA was ordered. If there are no objective connective tissue disease features, avoid diagnosing lupus from ANA alone. If features suggest SLE or related disease, obtain targeted tests: CBC, urinalysis, creatinine, complements, anti-dsDNA, anti-Sm, SSA/SSB, RNP, antiphospholipid panel if thrombosis/pregnancy morbidity, and disease-specific evaluation. If systemic sclerosis or myositis features exist, use disease-specific antibodies and organ screening rather than repeating ANA. Interpret titer and pattern as supportive, not definitive.

Algorithm D: suspected vasculitis

Identify vessel size and organ phenotype: skin purpura, kidney, lung, ENT, nerve, eye, GI, large-vessel ischemia. Exclude mimics: infection, endocarditis, emboli, drug reaction, malignancy, thrombotic disease, cholesterol emboli, and antiphospholipid syndrome. Obtain basic organ staging: CBC, CMP, urinalysis with microscopy, urine protein, ESR/CRP, chest imaging if pulmonary symptoms, hepatitis/HIV testing when relevant. Use targeted serology: ANCA PR3/MPO for pulmonary-renal/ENT/lung/nerve patterns, complements and cryoglobulins for immune-complex patterns, ANA/dsDNA for lupus phenotype. Biopsy the most accessible active organ when feasible because treatment may require high-risk immunosuppression.

High-yield integrative cases

Case 1: hot knee in a diabetic older adult. A 63-year-old has abrupt hot swollen knee, diabetes, and fever. The correct first branch is septic arthritis versus crystal arthritis, not "gout versus pseudogout." Aspirate the knee, culture the fluid, check crystals, and obtain blood cultures. If fluid is purulent or suspicion remains high, start antibiotics and drain. A prior gout history does not protect the joint from infection.

Case 2: early rheumatoid arthritis pattern. A 29-year-old has symmetric MCP/PIP swelling, morning stiffness for two hours, positive anti-CCP, and elevated CRP. This is an early RA pattern. The diagnostic weight comes from objective small-joint synovitis plus anti-CCP, not from joint pain alone. Early DMARD treatment is important because persistent synovitis predicts erosions and disability.

Case 3: lupus with nephritis risk. A 24-year-old has arthralgias, photosensitive rash, oral ulcers, anemia, proteinuria, low C3/C4, and positive anti-dsDNA. The joint pain is not the dangerous feature; nephritis is. Urinalysis, protein quantification, kidney function, complements, anti-dsDNA, and nephrology/rheumatology involvement are central. Pain control alone would miss organ-threatening lupus.

Case 4: vision-threatening giant cell arteritis. A 71-year-old has new headache, scalp tenderness, jaw claudication, ESR/CRP elevation, and transient visual blurring. Treat suspected giant cell arteritis immediately with high-dose glucocorticoids and obtain confirmatory temporal artery imaging or biopsy. Vision risk makes this a treatment-before-proof scenario.

Case 5: pulmonary-renal vasculitis phenotype. A 48-year-old has sinusitis, cavitary lung nodules, hematuria with RBC casts, and positive PR3-ANCA. This is a granulomatosis with polyangiitis phenotype until proven otherwise, but infection and malignancy remain important mimics. Kidney or other tissue biopsy is often pursued if feasible because induction therapy carries major risk.

Chapter summary

Rheumatology begins with clinical structure: articular versus periarticular, inflammatory versus mechanical, monoarticular versus polyarticular, acute versus chronic, axial versus peripheral, and localized versus systemic. Acute monoarthritis demands exclusion of septic arthritis. Polyarthritis should be mapped by joint distribution, symmetry, tempo, and extra-articular clues. Autoantibodies should refine a suspected syndrome rather than screen nonspecific pain. Complement patterns help identify immune-complex activity, especially in lupus, cryoglobulinemia, and infection-related glomerulonephritis. ESR and CRP measure inflammation but are not diagnostic. Synovial fluid is decisive in acute effusions but must be interpreted probabilistically. Vasculitis is organized by vessel size and organ threat. Immunosuppressive medications require infection prevention, vaccination planning, screening, and toxicity monitoring. The red flags that must not be missed are nephritis, pulmonary hemorrhage, CNS disease, vision loss, mononeuritis multiplex, digital ischemia, severe cytopenias, and hot swollen joints.

References and source framework

This chapter is original synthesized educational text. It uses the following clinical reference framework for accuracy and organization: American College of Rheumatology patient and guideline resources on rheumatoid arthritis, lupus, vasculitis, vaccination, and immunosuppressive treatment principles; Merck Manual Professional discussions of monoarticular and polyarticular joint pain and acute infectious arthritis; CDC guidance on altered immunocompetence and high-dose corticosteroid thresholds; and peer-reviewed ACR/Vasculitis Foundation and ACR vaccination guidelines. These sources were used for general clinical guardrails, not copied as chapter prose.

1. American College of Rheumatology. Rheumatoid Arthritis patient resource and rheumatoid arthritis treatment guideline resources.
2. American College of Rheumatology/Vasculitis Foundation. Clinical practice guidelines for vasculitis management.
3. American College of Rheumatology. Lupus and antiphospholipid syndrome patient/guideline resources.
4. Merck Manual Professional. Evaluation of joint symptoms; pain in and around a single joint; pain in multiple joints; acute infectious arthritis.
5. Bass AR, et al. 2022 American College of Rheumatology Guideline for Vaccinations in Patients With Rheumatic and Musculoskeletal Diseases. *Arthritis Care Res.* 2023.
6. Centers for Disease Control and Prevention. Altered immunocompetence and immunization best practices; immunocompromised vaccination guidance.